Time Series Agent: Components and Information Flow

Time series & context

observations · events · task request

LLM Side

§3

interprets temporal evidence and selects actions

- Representation alignment
- Temporal instruction & reasoning
- Workflow policy learning

e.g. ChatTS, Time-LLM, TimeRLM

action

General Harness

§4

executes actions, keeps state, coordinates roles

- Interface & Interaction
- Memory & Context
- Tools & Execution
- Planning & Orchestration
- Optimization & Evolution

e.g. TimeClaw-Harness, AION, TS-Agent

observation

feedback: revise

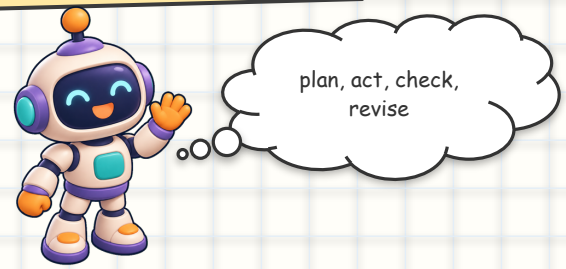
Validation & Evaluation

§7

checks process, temporal validity, task and decision utility

- Execution checks
- Information-cutoff checks
- Task metrics
- Downstream utility

e.g. TimeSeriesGym, Dr-CiK, TraceBench



temporal operations

General Time Series Harness

§5

supplies temporal data, tools and time constraints

- Temporal Data Management
- Temporal Processing
- Temporal Context & Memory
- Temporal Validation

e.g. TimeSage-EV, OpenForecaster, TSQAgent

Task-Specific Harness

§6

organizes outputs around task objectives

- Forecasting & Prediction
- Data Augmentation & Synthesis
- Anomaly Detection & Diagnosis
- Decision Support

e.g. FLAIRR-TS, MetaCaster, Argos

shared temporal capabilities

candidate output

Forecast · diagnosis · decision

accept / terminate

blue → green → red notes: scope of reuse: general → specific